

武器系统动力学建模与仿真

拉格朗日方程

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① 系统广义坐标与约束方程

② 虚功原理与拉格朗日方程

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系统广义坐标

Usually, n generalized coordinates, denoted as $q_j(t)$, $j = 1, 2, \dots, n$, are collected to describe the configuration, namely, the position of each particle, of a multibody system, which could also be denoted as a column vector

$$\mathbf{q}(t) = \begin{bmatrix} q_1(t) & q_2(t) & \dots & q_n(t) \end{bmatrix}^T.$$

For a particle numbered k , its global position may be formulated as

$$\mathbf{r}_k(q_1(t), q_2(t), \dots, q_n(t)) =: \mathbf{r}_k(\mathbf{q}(t)).$$

Among the various choices, the minimum number of n is called the degrees of freedom (DOFs) of the multibody system.

系统广义速度

Taking the first time derivative of

$$\mathbf{r}_k(q_1(t), q_2(t), \dots, q_n(t)) =: \mathbf{r}_k(\mathbf{q}(t))$$

with respect to time leads to the global absolute velocity of partical k

$$\begin{aligned}\frac{d}{dt}\mathbf{r}_k(\mathbf{q}(t)) &= \sum_{i=1}^n \frac{\partial \mathbf{r}_k}{\partial q_i} \frac{dq_i}{dt} \\ &= \sum_{i=1}^n \frac{\partial \mathbf{r}_k}{\partial q_i} \dot{q}_i \\ &=: \dot{\mathbf{r}}_k(\mathbf{q}(t), \dot{\mathbf{q}}(t)).\end{aligned}$$

The combined set of the generalized coordinates $\mathbf{q}(t)$ of the system and the generalized velocities $\dot{\mathbf{q}}(t)$ are called the **state variables of the whole system**.

系统广义坐标的约束方程

Generalized coordinates of a multibody system

$$\mathbf{q}(t) = \begin{bmatrix} q_1(t) & q_2(t) & \dots & q_n(t) \end{bmatrix}^T.$$

If the number of the generalized coordinates is greater than the DOFs of the system, there will be corresponding constraint equations imposed on the generalized coordinates, denoted as

$$0 = c_i(\mathbf{q}(t), t) \quad (i = 1, 2, \dots, m).$$

These constraint equations may be denoted in a compact form as

$$\mathbf{0} = \mathbf{c}(\mathbf{q}(t), t),$$

where

$$\mathbf{c}(\mathbf{q}(t), t) = \begin{bmatrix} c_1(\mathbf{q}(t), t) & c_2(\mathbf{q}(t), t) & \dots & c_m(\mathbf{q}(t), t) \end{bmatrix}^T.$$

约束方程对时间的导数

First time derivative of constraint equation $0 = c_i(\mathbf{q}(t), t)$ reads as

$$0 = \sum_{j=1}^n \frac{\partial c_i}{\partial q_j} \dot{q}_j + \frac{\partial c_i}{\partial t} =: \dot{c}_i(\mathbf{q}(t), \dot{\mathbf{q}}(t), t) \quad (i = 1, 2, \dots, m).$$

These velocity-level constraint equations may be collected and denoted in a compact form as

$$\mathbf{0} = \mathbf{c}_q(\mathbf{q}(t), t) \dot{\mathbf{q}}(t) + \mathbf{c}_t(\mathbf{q}(t), t),$$

where

$$\mathbf{c}_q(\mathbf{q}(t), t) = \begin{bmatrix} \frac{\partial c_1}{\partial q_1} & \frac{\partial c_1}{\partial q_2} & \cdots & \frac{\partial c_1}{\partial q_n} \\ \frac{\partial c_2}{\partial q_1} & \frac{\partial c_2}{\partial q_2} & \cdots & \frac{\partial c_2}{\partial q_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial c_m}{\partial q_1} & \frac{\partial c_m}{\partial q_2} & \cdots & \frac{\partial c_m}{\partial q_n} \end{bmatrix}, \quad \mathbf{c}_t(\mathbf{q}(t), t) = \begin{bmatrix} \frac{\partial c_1}{\partial t} \\ \frac{\partial c_2}{\partial t} \\ \vdots \\ \frac{\partial c_m}{\partial t} \end{bmatrix}.$$

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质点的运动微分方程

Newton's second law of motion says that

$$\mathbf{f} = \frac{d}{dt}(m\mathbf{v}) = \frac{d}{dt}\left(m\frac{d\mathbf{r}}{dt}\right) = \frac{dm}{dt}\frac{d\mathbf{r}}{dt} + m\frac{d^2\mathbf{r}}{dt^2}.$$

For a partical with constant mass, there is its concise form denoted as

$$\mathbf{f} = m\frac{d^2\mathbf{r}}{dt^2} =: m\ddot{\mathbf{r}} =: m\mathbf{a}.$$

Treating $(-m\ddot{\mathbf{r}})$ as an inertial force leads to the D'Alembert principle

$$-m\ddot{\mathbf{r}} + \mathbf{f} = \mathbf{0}.$$

For partical k involved in a multibody system, there is

$$-m_k\ddot{\mathbf{r}}_k + \mathbf{f}_{\text{Ex},k} + \mathbf{f}_{\text{In},k} = \mathbf{0}.$$

广义坐标的微小变更

Assume that, at an arbitrary time instant t , the configuration of the system is specified by a set of generalized coordinates $\mathbf{q}(t)$, which are confined by and consistently complied to the constraint equations

$$0 = c_i(\mathbf{q}(t), t) =: c_i(\mathbf{q}, t) \quad (i = 1, 2, \dots, m).$$

At the very moment of time instant t , explore possible instantaneous displacement determined by very small increment of $\mathbf{q}(t)$, denoted as $\delta\mathbf{q}(t)$, which are called **generalized virtual displacement of the system**, rather than the actual one. Of course, the generalized virtual displacements $\delta\mathbf{q}(t)$ are complied to the constraint equations, namely

$$0 = c_i(\mathbf{q}(t) + \delta\mathbf{q}(t), t) =: c_i(\mathbf{q} + \delta\mathbf{q}, t) \quad 0 < |\delta\mathbf{q}| \ll 1.$$

广义坐标的微小变更

Based on

$$0 = c_i(\mathbf{q}, t) \quad (i = 1, 2, \dots, m)$$

and

$$0 = c_i(\mathbf{q} + \delta\mathbf{q}, t) \quad 0 < |\delta\mathbf{q}| \ll 1,$$

application of the first-order Taylor expansion results in

$$0 = c_i(\mathbf{q} + \delta\mathbf{q}, t) \approx c_i(\mathbf{q}, t) + \sum_{j=1}^n \frac{\partial c_i}{\partial q_j} \delta q_j = \sum_{j=1}^n \frac{\partial c_i}{\partial q_j} \delta q_j.$$

广义坐标的微小变更

Combining all the constraint equations

$$0 = \sum_{j=1}^n \frac{\partial c_i}{\partial q_j} \delta q_j \quad (i = 1, 2, \dots, m)$$

leads to

$$\begin{aligned} \mathbf{0} &= \begin{bmatrix} \frac{\partial c_1}{\partial q_1} & \frac{\partial c_1}{\partial q_2} & \cdots & \frac{\partial c_1}{\partial q_n} \\ \frac{\partial c_2}{\partial q_1} & \frac{\partial c_2}{\partial q_2} & \cdots & \frac{\partial c_2}{\partial q_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial c_m}{\partial q_1} & \frac{\partial c_m}{\partial q_2} & \cdots & \frac{\partial c_m}{\partial q_n} \end{bmatrix} \begin{bmatrix} \delta q_1 \\ \delta q_2 \\ \vdots \\ \delta q_n \end{bmatrix} \\ &=: \mathbf{c}_q(\mathbf{q}, t) \delta \mathbf{q}, \end{aligned}$$

where $\mathbf{c}_q$ is called Jacobian matrix of the constraint equations.

质点的虚位移

Global position vector of partical k represented by the generalized coordinates reads as

$$\mathbf{r}_k = \mathbf{r}_k(\mathbf{q}(t)).$$

Its instantaneous minor change caused by the generalized virtual displacement $\delta\mathbf{q}$ satisfies

$$\delta\mathbf{r}_k = \mathbf{r}_k(\mathbf{q} + \delta\mathbf{q}) - \mathbf{r}_k(\mathbf{q}) \approx \sum_{j=1}^n \frac{\partial \mathbf{r}_k}{\partial q_j} \delta q_j,$$

which is called **the virtual displacement of partical k** .

质点与质点系的虚功方程

Inner product of the virtual displacement of partical k , namely $\delta \mathbf{q}$, and its dynamics equation

$$-m_k \ddot{\mathbf{r}}_k + \mathbf{f}_{\text{Ex},k} + \mathbf{f}_{\text{In},k} = \mathbf{0}$$

leads to its virtual work equation, namely

$$-m_k \delta \mathbf{r}_k^T \ddot{\mathbf{r}}_k + \delta \mathbf{r}_k^T \mathbf{f}_{\text{Ex},k} + \delta \mathbf{r}_k^T \mathbf{f}_{\text{In},k} = 0.$$

Summation with respect to partical index k leads to the virtual work equation of the whole system, namely

$$-\sum_{k=1}^N m_k \delta \mathbf{r}_k^T \ddot{\mathbf{r}}_k + \sum_{k=1}^N \delta \mathbf{r}_k^T \mathbf{f}_{\text{Ex},k} + \sum_{k=1}^N \delta \mathbf{r}_k^T \mathbf{f}_{\text{In},k} = 0.$$

系统惯性力的虚功

For the first term, namely the virtual work of the inertia forces,

$$\begin{aligned} -\sum_{k=1}^N m_k \delta \mathbf{r}_k^T \ddot{\mathbf{r}}_k &= -\sum_{k=1}^N m_k \left(\sum_{j=1}^n \frac{\partial \mathbf{r}_k}{\partial q_j} \delta q_j \right)^T \frac{d}{dt} (\dot{\mathbf{r}}_k) \\ &= -\sum_{k=1}^N m_k \sum_{j=1}^n \left(\frac{\partial \mathbf{r}_k}{\partial q_j} \delta q_j \right)^T \frac{d}{dt} (\dot{\mathbf{r}}_k) \\ &= -\sum_{k=1}^N m_k \sum_{j=1}^n \delta q_j \frac{\partial \mathbf{r}_k^T}{\partial q_j} \frac{d}{dt} (\dot{\mathbf{r}}_k) \\ &= -\sum_{k=1}^N \sum_{j=1}^n m_k \delta q_j \frac{\partial \mathbf{r}_k^T}{\partial q_j} \frac{d}{dt} (\dot{\mathbf{r}}_k) \end{aligned}$$

系统惯性力的虚功

The virtual work of the inertia forces

$$\begin{aligned} -\sum_{k=1}^N m_k \delta \mathbf{r}_k^T \ddot{\mathbf{r}}_k &= -\sum_{k=1}^N \sum_{j=1}^n m_k \delta q_j \frac{\partial \mathbf{r}_k^T}{\partial q_j} \frac{d}{dt} (\dot{\mathbf{r}}_k) \\ &= -\sum_{j=1}^n \sum_{k=1}^N m_k \delta q_j \frac{\partial \mathbf{r}_k^T}{\partial q_j} \frac{d}{dt} (\dot{\mathbf{r}}_k) \\ &= -\sum_{j=1}^n \delta q_j \sum_{k=1}^N m_k \left[\left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \right) \frac{d}{dt} (\dot{\mathbf{r}}_k) \right] \\ &= -\sum_{j=1}^n \delta q_j \sum_{k=1}^N m_k \left[\frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \dot{\mathbf{r}}_k \right) - \frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \right) \dot{\mathbf{r}}_k \right]. \end{aligned}$$

系统惯性力的虚功

The virtual work of the inertia forces

$$\begin{aligned} -\sum_{k=1}^N m_k \delta \mathbf{r}_k^T \ddot{\mathbf{r}}_k &= -\sum_{j=1}^n \delta q_j \sum_{k=1}^N m_k \left[\frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \dot{\mathbf{r}}_k \right) - \frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \right) \dot{\mathbf{r}}_k \right] \\ &= -\sum_{j=1}^n \delta q_j \left[\sum_{k=1}^N m_k \frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \dot{\mathbf{r}}_k \right) - \sum_{k=1}^N m_k \frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \right) \dot{\mathbf{r}}_k \right] \\ &= -\sum_{j=1}^n \delta q_j \left[\frac{d}{dt} \left(\sum_{k=1}^N m_k \frac{\partial \mathbf{r}_k^T}{\partial q_j} \dot{\mathbf{r}}_k \right) - \sum_{k=1}^N m_k \frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \right) \dot{\mathbf{r}}_k \right] \end{aligned}$$

系统惯性力的虚功

Note that

$$\frac{\partial \dot{\mathbf{r}}_k}{\partial \dot{q}_j} = \frac{\partial \mathbf{r}_k}{\partial q_j}, \quad \frac{d}{dt} \left(\frac{\partial \mathbf{r}_k}{\partial q_j} \right) = \frac{\partial \dot{\mathbf{r}}_k}{\partial q_j}.$$

Proof tips:

$$\dot{\mathbf{r}}_k = \frac{d}{dt} \mathbf{r}_k(\mathbf{q}(t)) = \sum_{i=1}^n \frac{\partial \mathbf{r}_k}{\partial q_i} \frac{dq_i}{dt} = \sum_{i=1}^n \frac{\partial \mathbf{r}_k}{\partial q_i} \dot{q}_i,$$

$$\frac{\partial \dot{\mathbf{r}}_k}{\partial q_j} = \sum_{i=1}^n \frac{\partial^2 \mathbf{r}_k}{\partial q_i \partial q_j} \dot{q}_i,$$

$$\frac{d}{dt} \left(\frac{\partial \mathbf{r}_k}{\partial q_j} \right) = \sum_{i=1}^n \frac{\partial}{\partial q_i} \left(\frac{\partial \mathbf{r}_k}{\partial q_j} \right) \dot{q}_i = \sum_{i=1}^n \frac{\partial^2 \mathbf{r}_k}{\partial q_j \partial q_i} \dot{q}_i.$$

系统惯性力的虚功: $-\sum_{k=1}^N m_k \delta \mathbf{r}_k^T \ddot{\mathbf{r}}_k =$
 $-\sum_{j=1}^n \delta q_j \left[\frac{d}{dt} \left(\sum_{k=1}^N m_k \frac{\partial \mathbf{r}_k^T}{\partial \dot{q}_j} \dot{\mathbf{r}}_k \right) - \sum_{k=1}^N m_k \frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial \dot{q}_j} \right) \dot{\mathbf{r}}_k \right]$

$$\begin{aligned} \sum_{k=1}^N m_k \frac{\partial \mathbf{r}_k^T}{\partial \dot{q}_j} \dot{\mathbf{r}}_k &= \sum_{k=1}^N m_k \frac{\partial \dot{\mathbf{r}}_k^T}{\partial \dot{q}_j} \dot{\mathbf{r}}_k \\ &= \sum_{k=1}^N m_k \frac{\partial}{\partial \dot{q}_j} \left(\frac{1}{2} \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k \right) \\ &= \frac{\partial}{\partial \dot{q}_j} \left(\sum_{k=1}^N \frac{1}{2} m_k \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k \right) \\ &=: \frac{\partial}{\partial \dot{q}_j} T(\mathbf{q}(t), \dot{\mathbf{q}}(t)). \end{aligned}$$

系统惯性力的虚功: $-\sum_{k=1}^N m_k \delta \mathbf{r}_k^T \ddot{\mathbf{r}}_k =$
 $-\sum_{j=1}^n \delta q_j \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \sum_{k=1}^N m_k \frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \right) \dot{\mathbf{r}}_k \right]$

$$\begin{aligned} \sum_{k=1}^N m_k \frac{d}{dt} \left(\frac{\partial \mathbf{r}_k^T}{\partial q_j} \right) \dot{\mathbf{r}}_k &= \sum_{k=1}^N m_k \frac{\partial \dot{\mathbf{r}}_k^T}{\partial q_j} \dot{\mathbf{r}}_k \\ &= \sum_{k=1}^N m_k \frac{\partial}{\partial q_j} \left(\frac{1}{2} \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k \right) \\ &= \frac{\partial}{\partial q_j} \left(\sum_{k=1}^N m_k \frac{1}{2} \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k \right) \\ &= \frac{\partial T}{\partial q_j}. \end{aligned}$$

系统惯性力的虚功

$$\begin{aligned} -\sum_{k=1}^N m_k \delta \mathbf{r}_k^T \ddot{\mathbf{r}}_k &= -\sum_{j=1}^n \delta q_j \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} \right] \\ &= \sum_{j=1}^n \delta q_j \left[-\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) + \frac{\partial T}{\partial q_j} \right]. \end{aligned}$$

系统外力与内力的虚功

For the virtual work of external and internal forces

$$\begin{aligned} \sum_{k=1}^N \delta \mathbf{r}_k^T \mathbf{f}_{\text{Ex},k} + \sum_{k=1}^N \delta \mathbf{r}_k^T \mathbf{f}_{\text{In},k} &= \sum_{k=1}^N \delta \mathbf{r}_k^T \mathbf{f}_{\text{Ex},k} + \sum_{k=1}^N \delta \mathbf{r}_k^T \mathbf{f}_{\text{InNonIdeal},k} \\ &= \sum_{k=1}^N \delta \mathbf{r}_k^T (\mathbf{f}_{\text{Ex},k} + \mathbf{f}_{\text{InNonIdeal},k}) \\ &= \sum_{k=1}^N \sum_{j=1}^n \delta q_j \frac{\partial \mathbf{r}_k^T}{\partial q_j} (\mathbf{f}_{\text{Ex},k} + \mathbf{f}_{\text{InNonIdeal},k}) \\ &= \sum_{j=1}^n \delta q_j \sum_{k=1}^N \frac{\partial \mathbf{r}_k^T}{\partial q_j} (\mathbf{f}_{\text{Ex},k} + \mathbf{f}_{\text{InNonIdeal},k}) \\ &=: \sum_{j=1}^n \delta q_j Q_j. \end{aligned}$$

拉格朗日方程

The virtual work equation is then simplified as

$$0 = \sum_{j=1}^n \delta q_j \left[-\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) + \frac{\partial T}{\partial q_j} + Q_j \right],$$

denoted in matrix form as

$$0 = \delta \mathbf{q}^T \left[-\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right)^T + \left(\frac{\partial T}{\partial \mathbf{q}} \right)^T + \mathbf{Q} \right],$$

which holds for any small generalized virtual displacement $\delta \mathbf{q}$ that is subject to the constraint equations

$$\mathbf{0} = \mathbf{c}_q(\mathbf{q}, t) \delta \mathbf{q}.$$

拉格朗日方程

Virtual work equation

$$0 = \delta \mathbf{q}^T \left[-\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right)^T + \left(\frac{\partial T}{\partial \mathbf{q}} \right)^T + \mathbf{Q} \right]$$

holds for any small $\delta \mathbf{q}$ subject to

$$\mathbf{0} = \mathbf{c}_q(\mathbf{q}, t) \delta \mathbf{q}.$$

Generalized variation principles says that there exists $\boldsymbol{\lambda} \in \mathbb{R}^m$ such that

$$-\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right)^T + \left(\frac{\partial T}{\partial \mathbf{q}} \right)^T + \mathbf{Q} - \mathbf{c}_q^T \boldsymbol{\lambda} = \mathbf{0},$$

which may also be denoted as

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right)^T - \left(\frac{\partial T}{\partial \mathbf{q}} \right)^T + \mathbf{c}_q^T \boldsymbol{\lambda} = \mathbf{Q}.$$